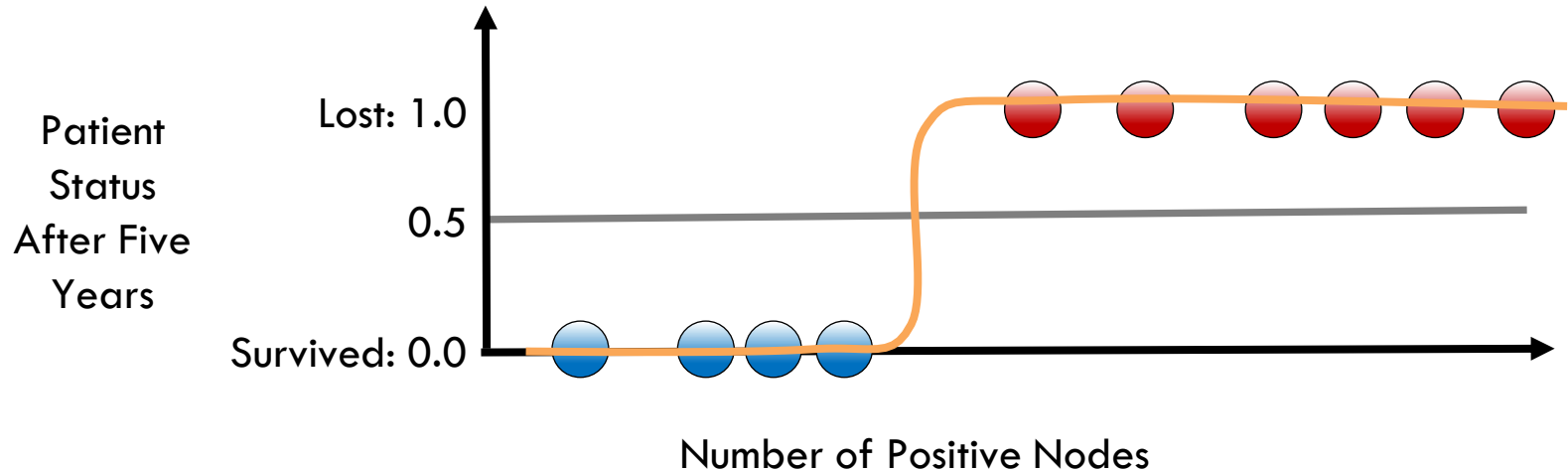

Support Vector Machines

Learning Objectives

- Apply support vector machines (SVMs)—a popular algorithm used for classification problems
- Recognize SVM similarity to logistic regression
- Apply regularization in SVMs and some tips to obtain non-linear classifications with SVMs

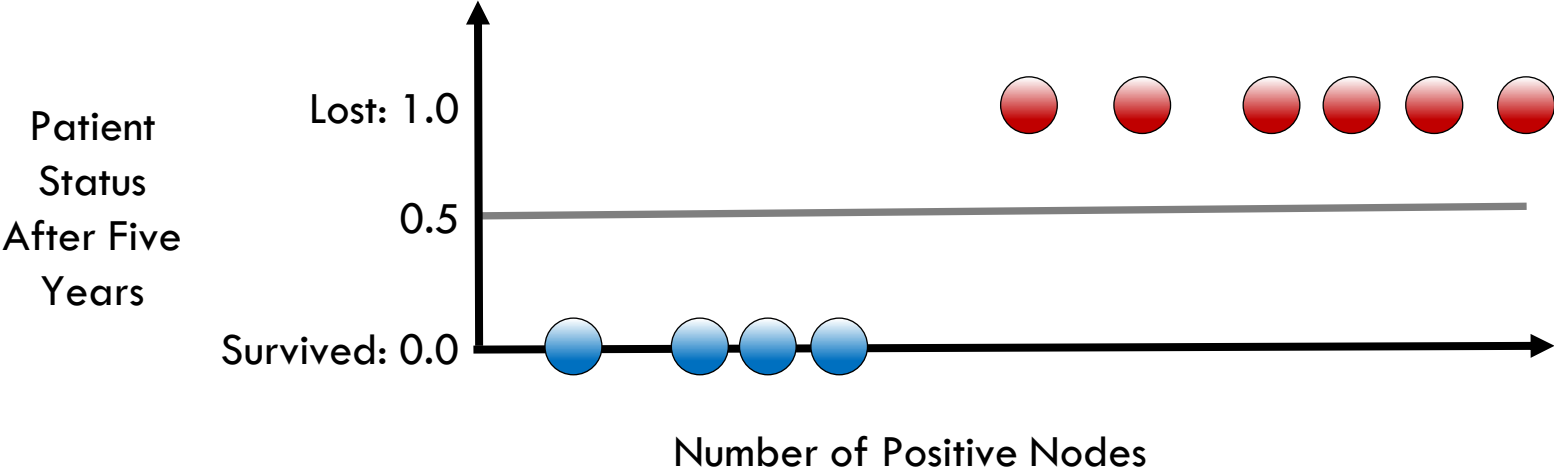


Relationship to Logistic Regression

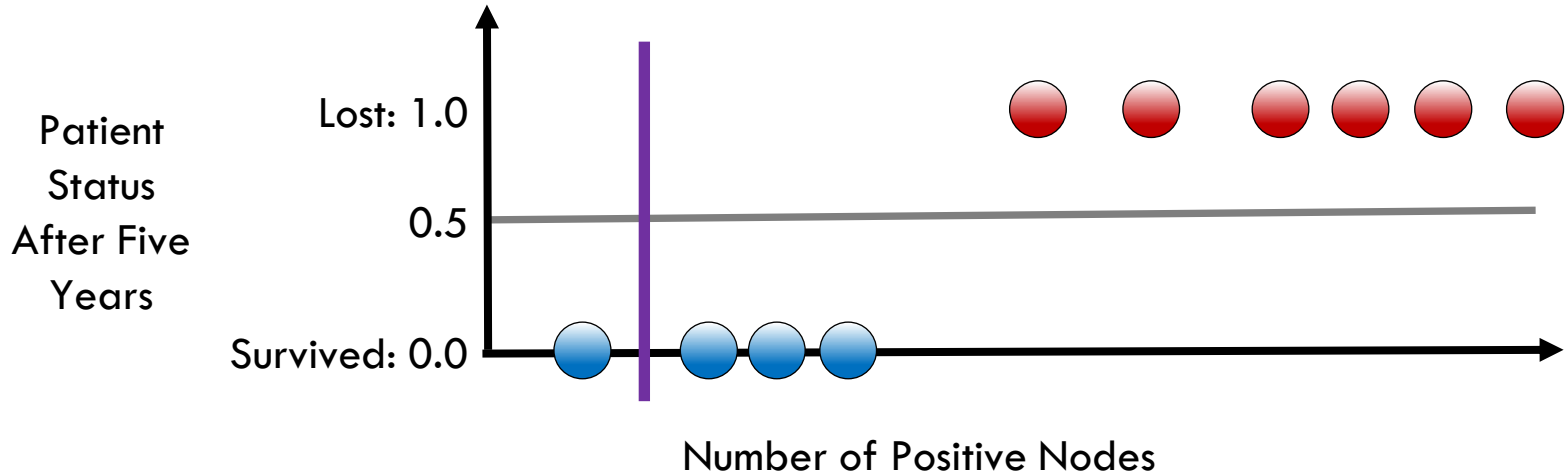


$$y_{\beta}(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x + \varepsilon)}}$$

Support Vector Machines (SVM)

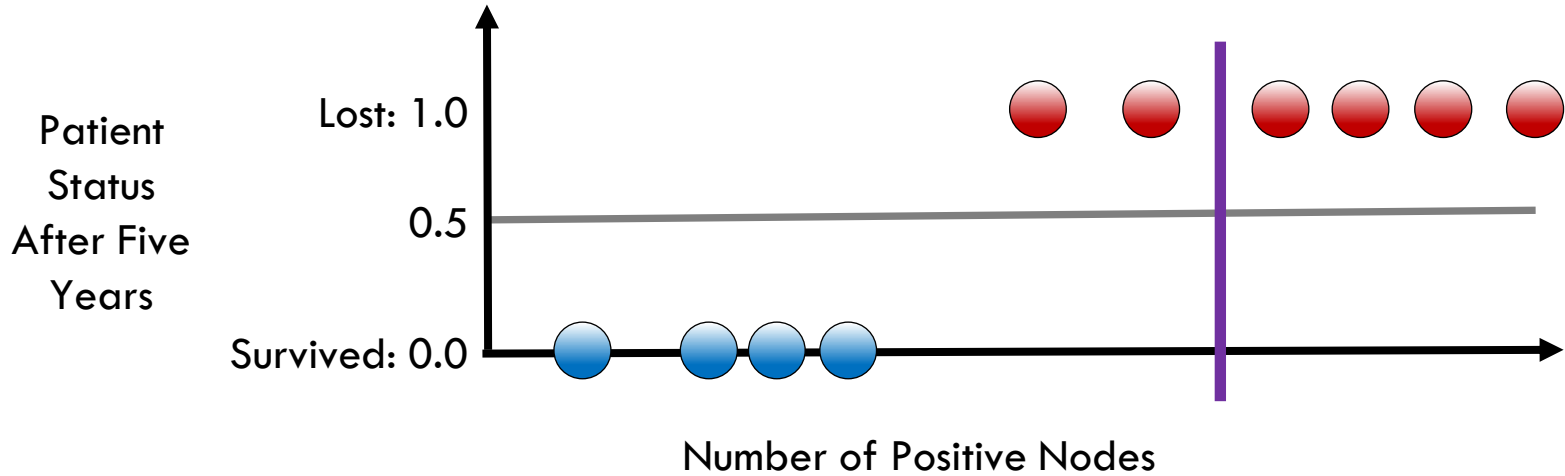


Support Vector Machines (SVM)



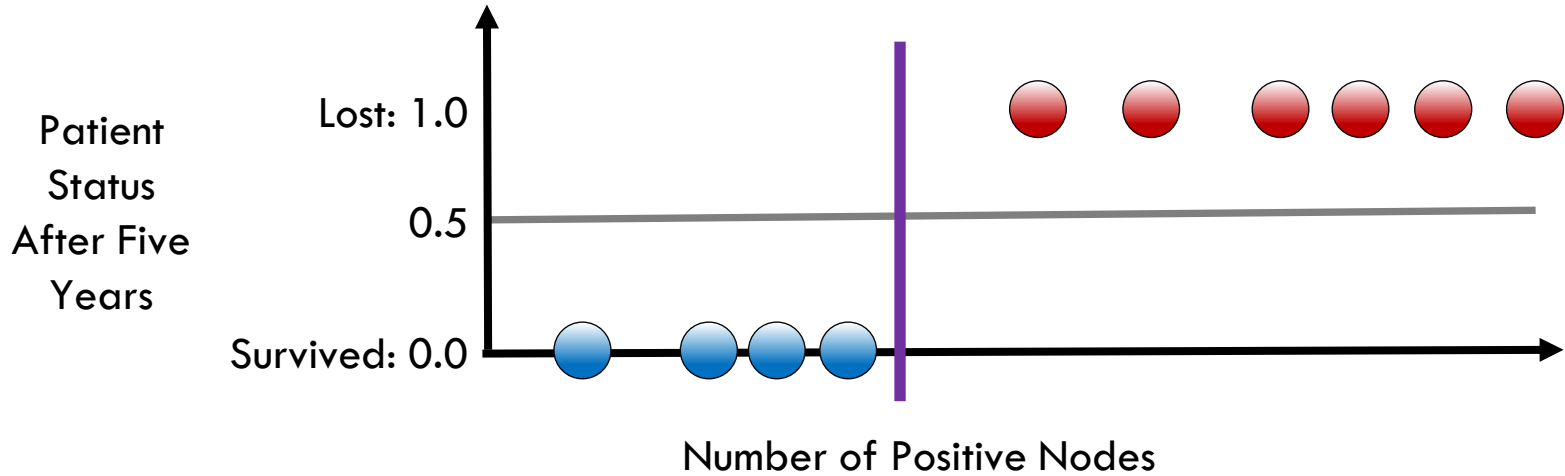
Three misclassifications

Support Vector Machines (SVM)



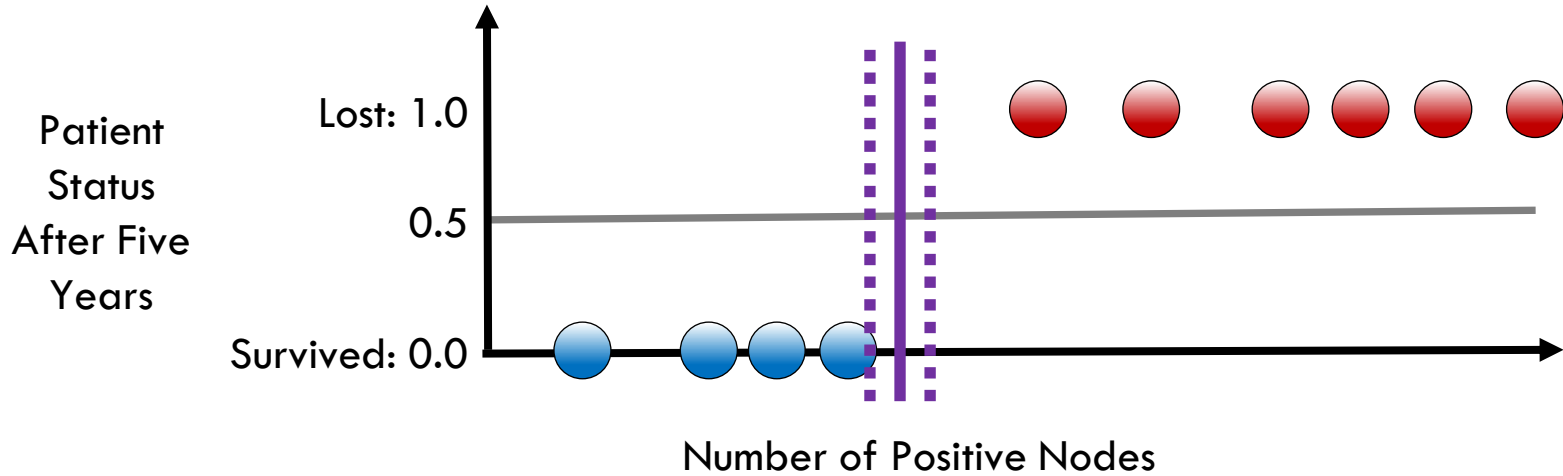
Two misclassifications

Support Vector Machines (SVM)



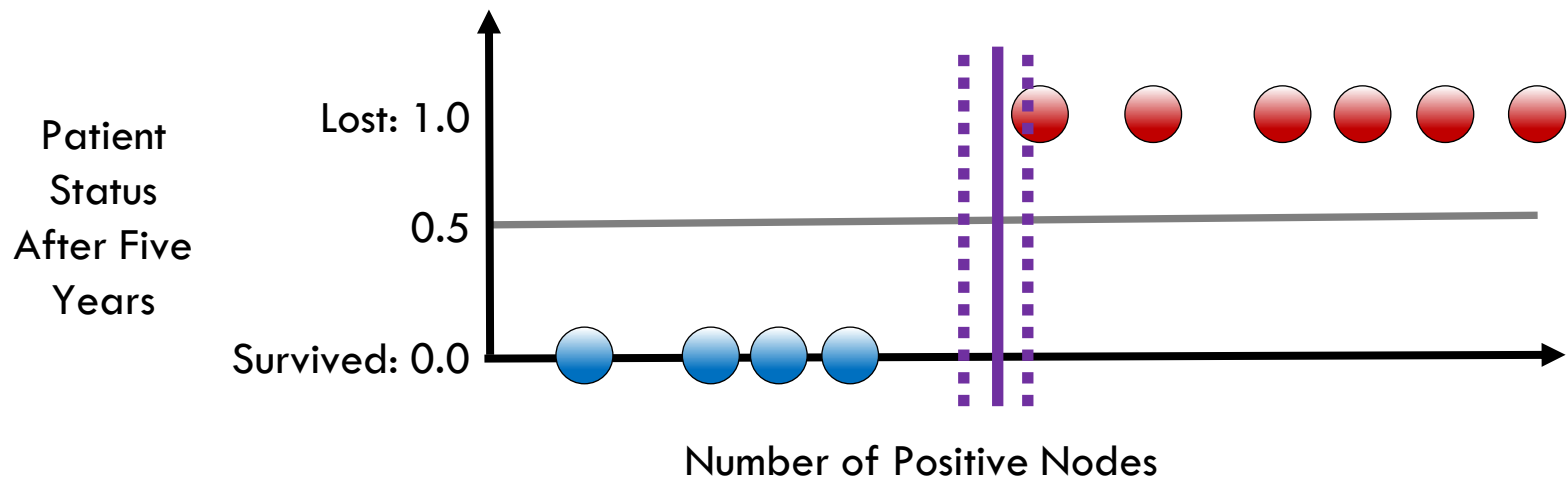
No misclassifications

Support Vector Machines (SVM)



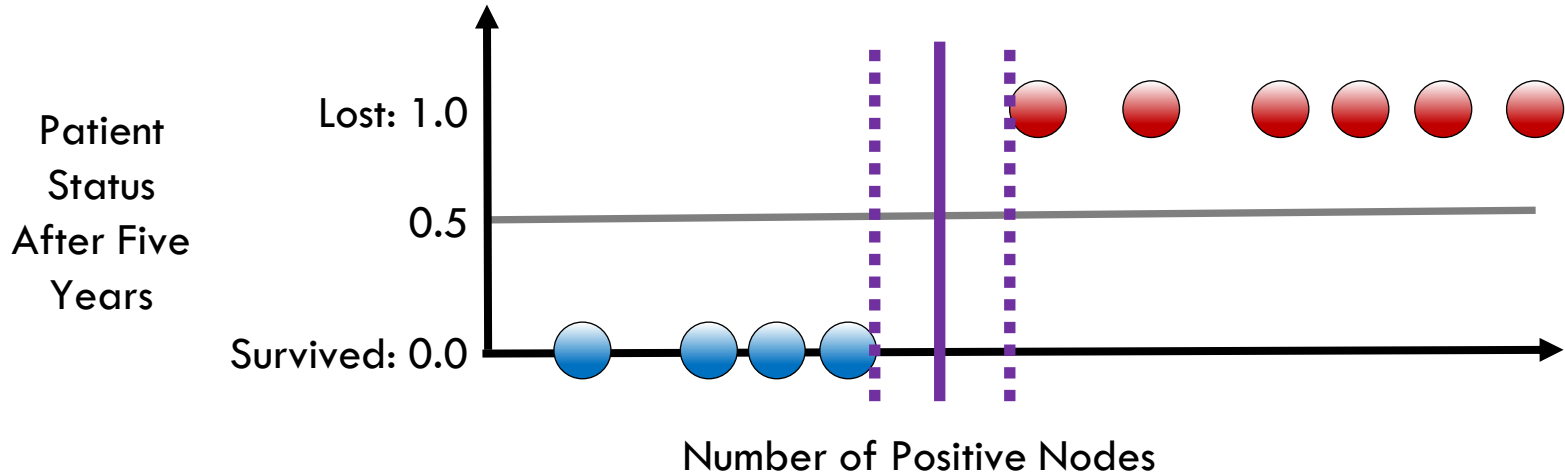
No misclassifications—but is this the best position?

Support Vector Machines (SVM)



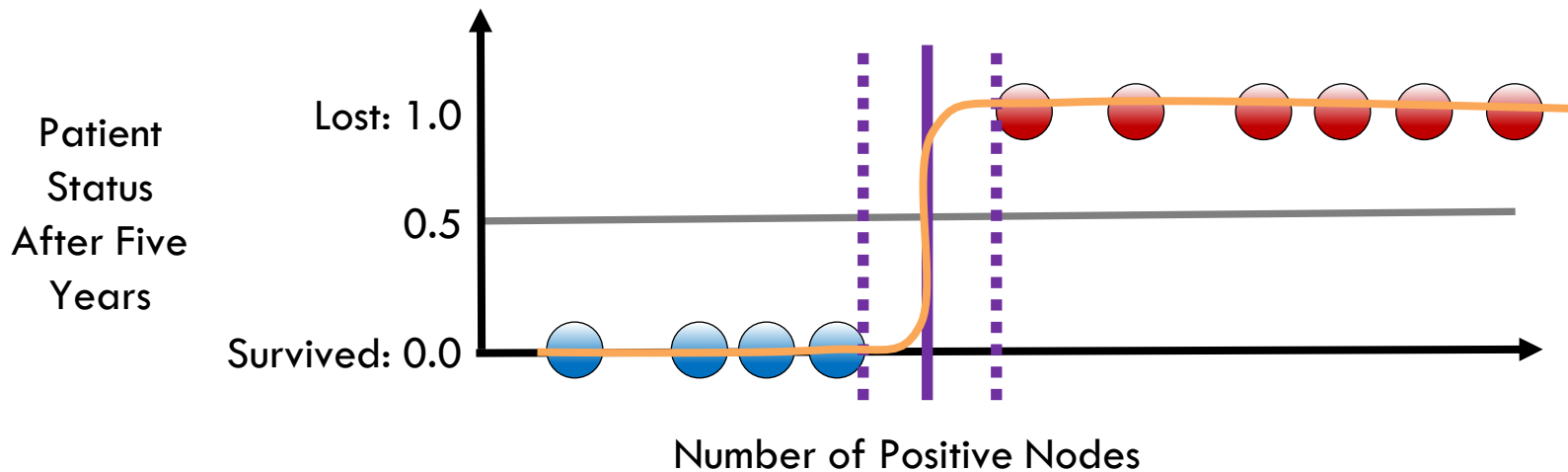
No misclassifications—but is this the best position?

Support Vector Machines (SVM)



Maximize the region between classes

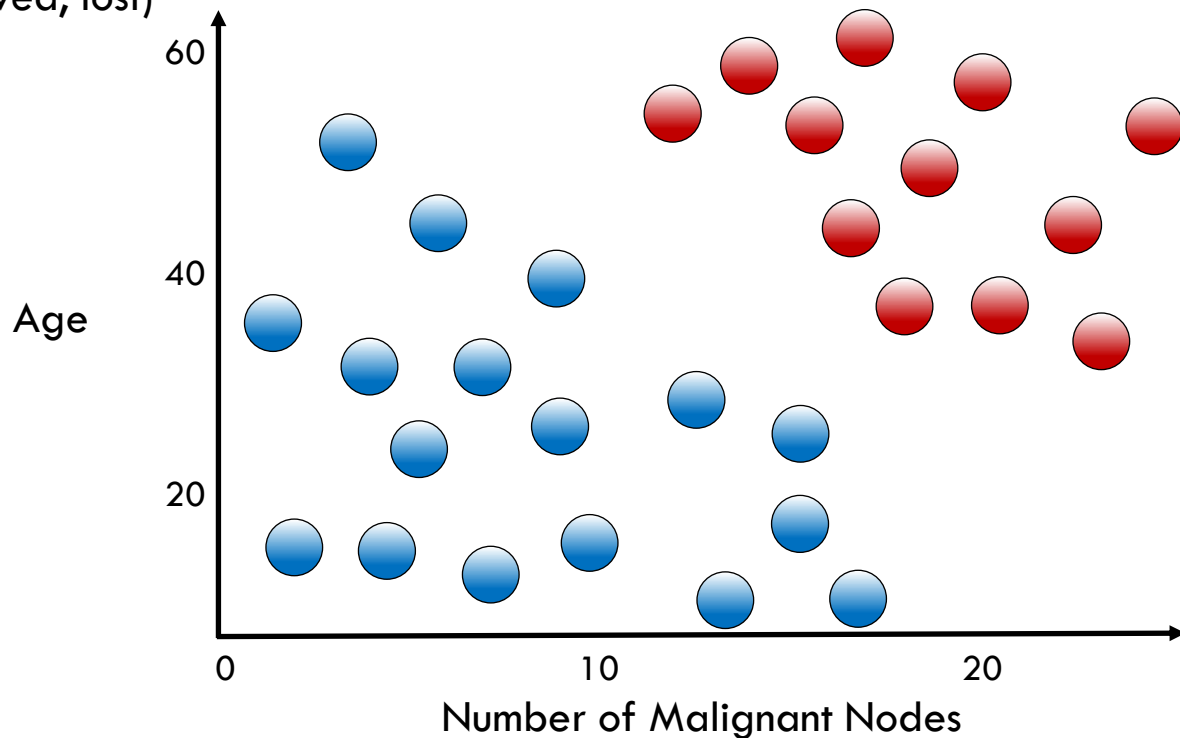
Similarity Between Logistic Regression and SVM



Classification with SVMs

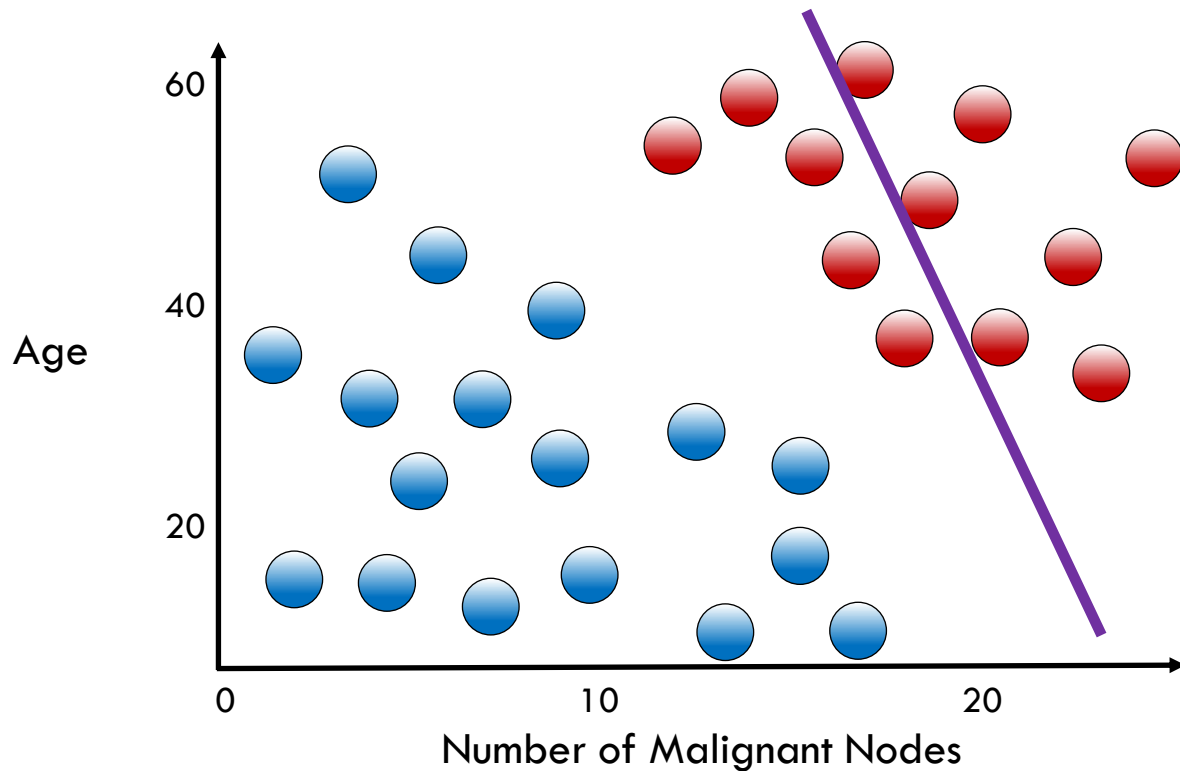
Two features (nodes, age)

Two labels (survived, lost)



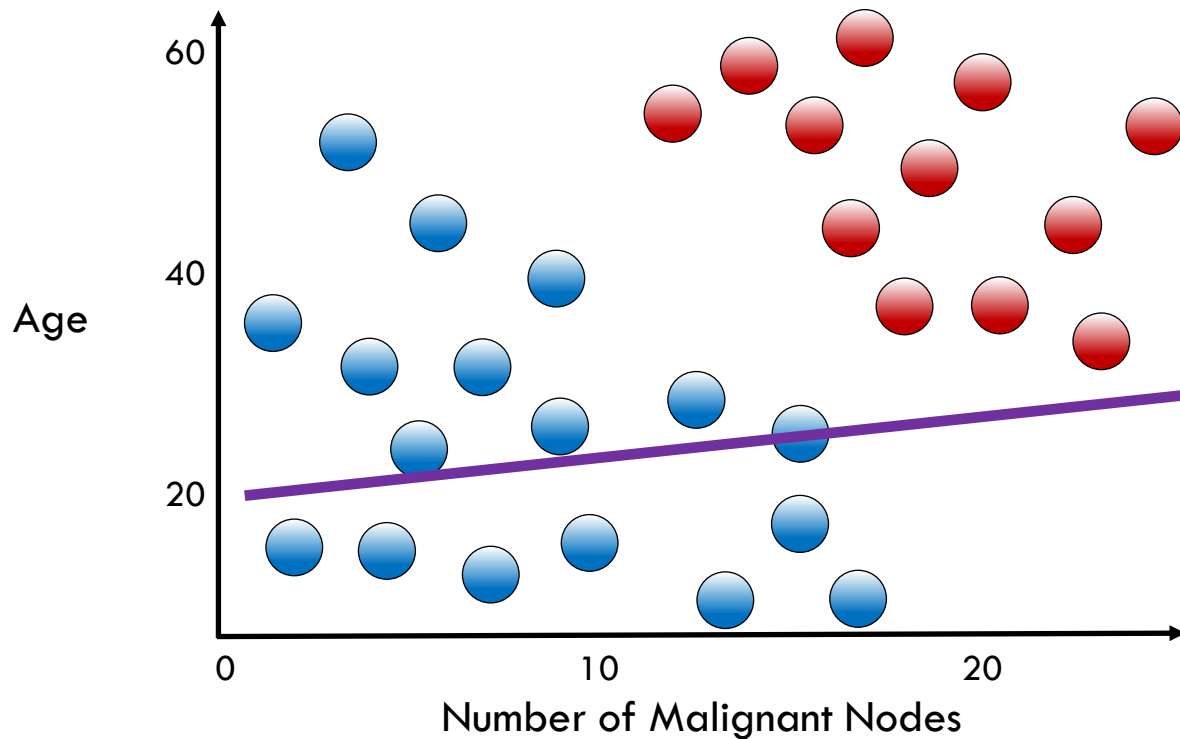
Classification with SVMs

Find the line that best separates
classes



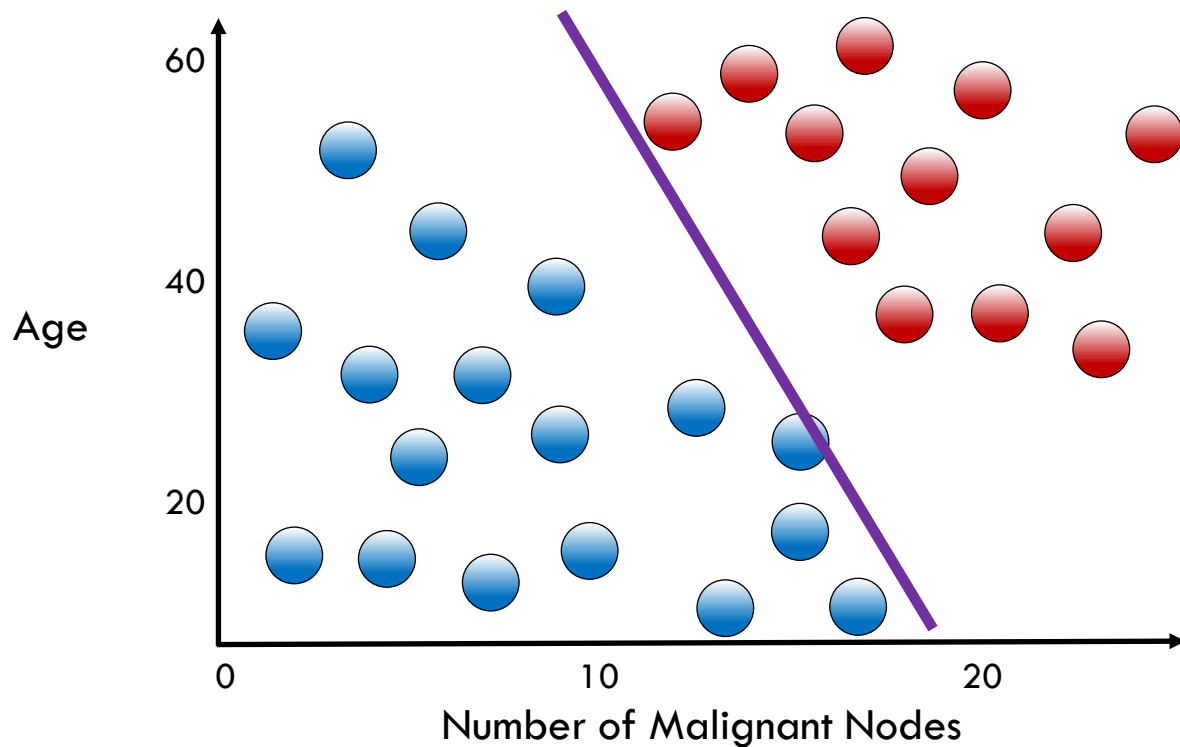
Classification with SVMs

Find the line that best separates
classes



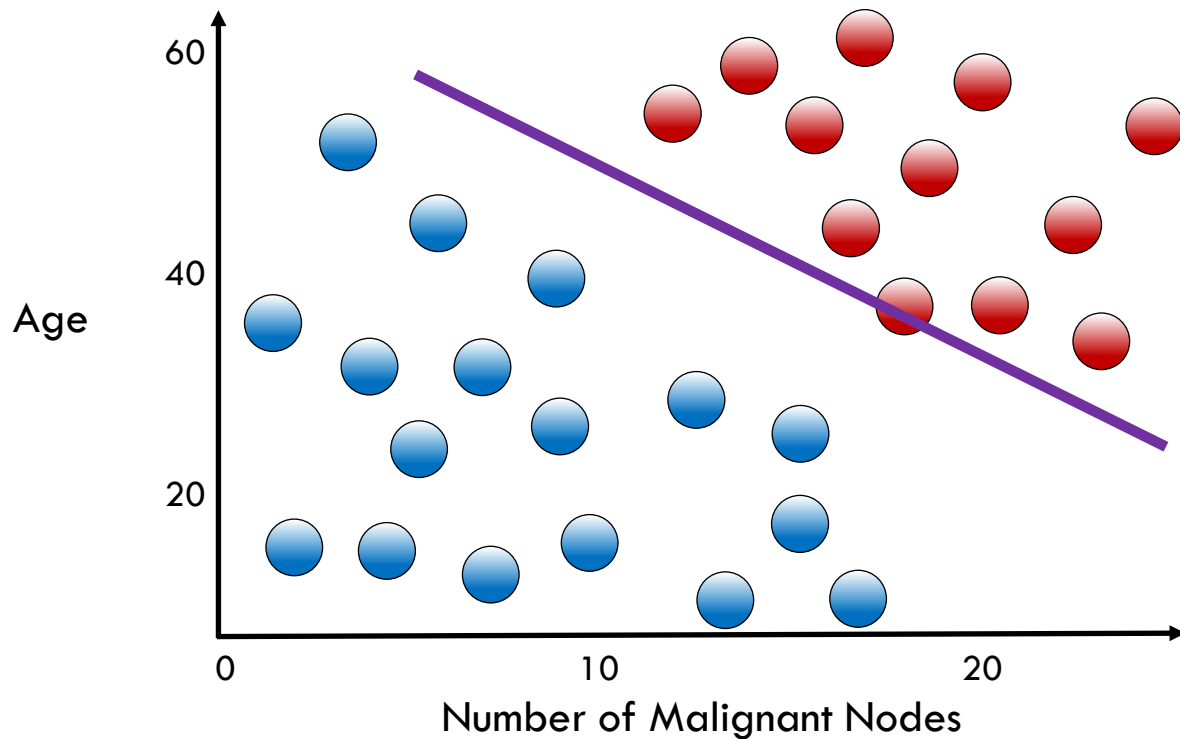
Classification with SVMs

Find the line that best separates
classes



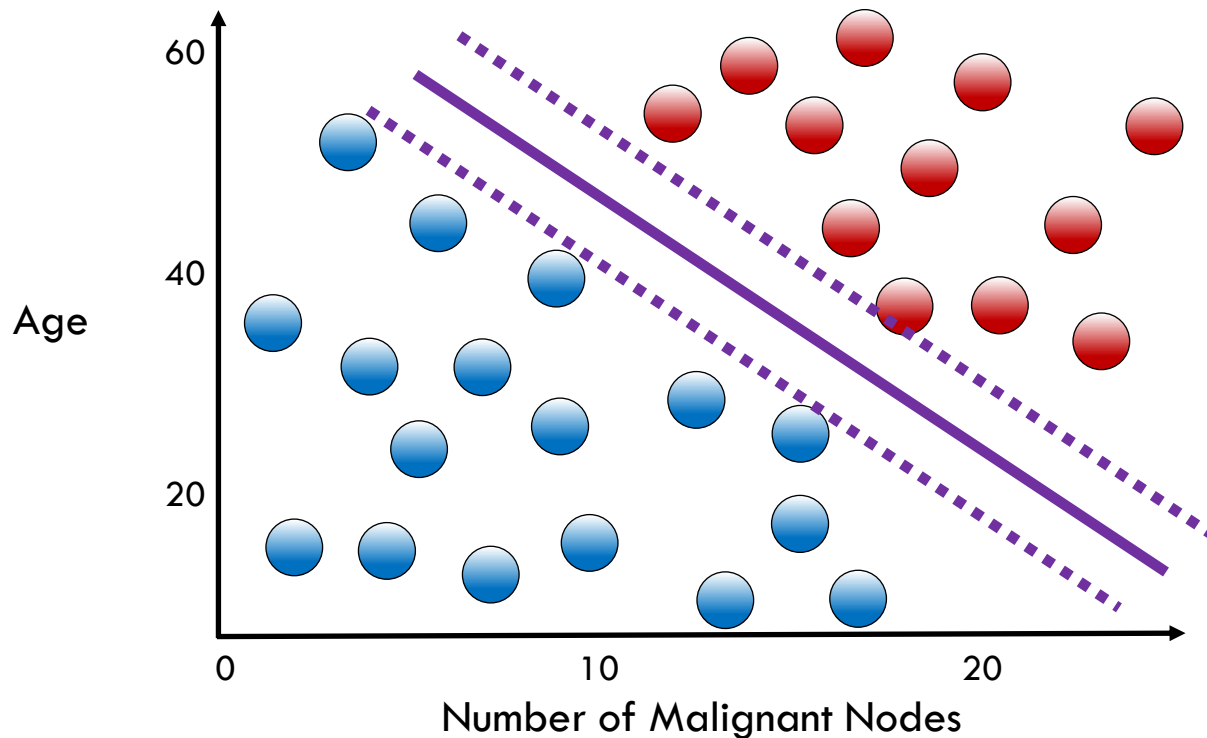
Classification with SVMs

Find the line that best separates
classes



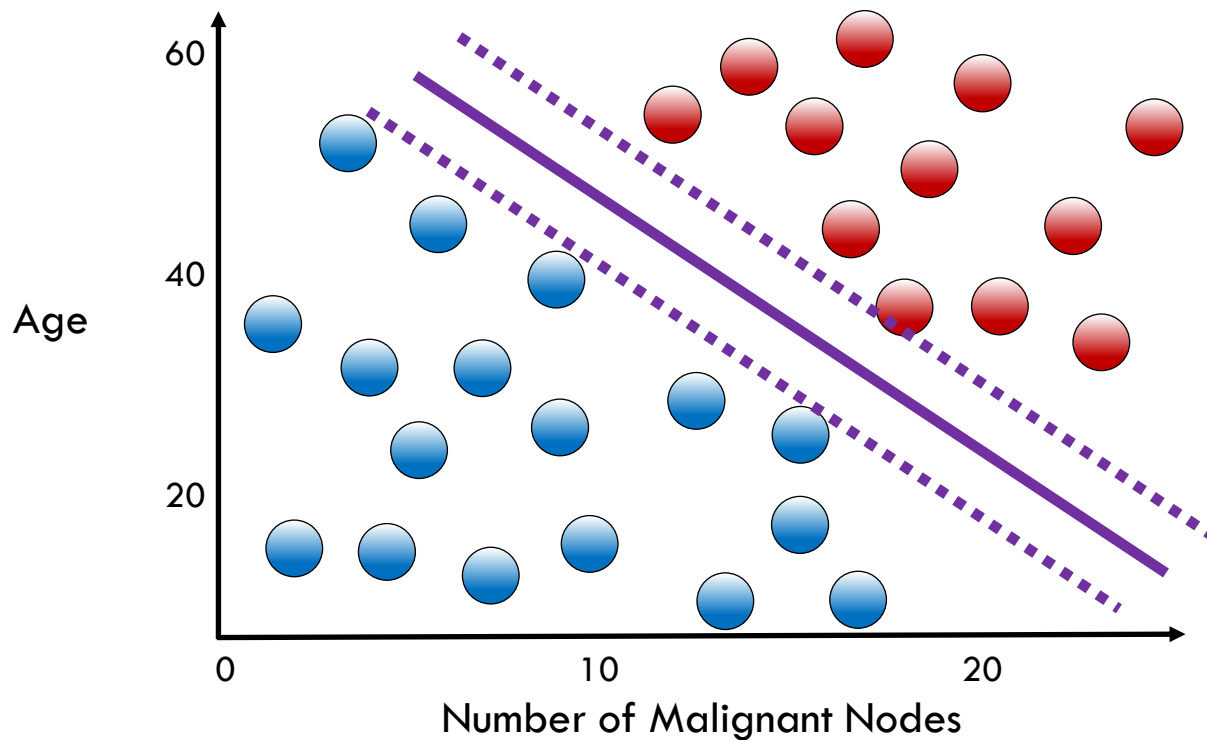
Classification with SVMs

And include the largest boundary possible



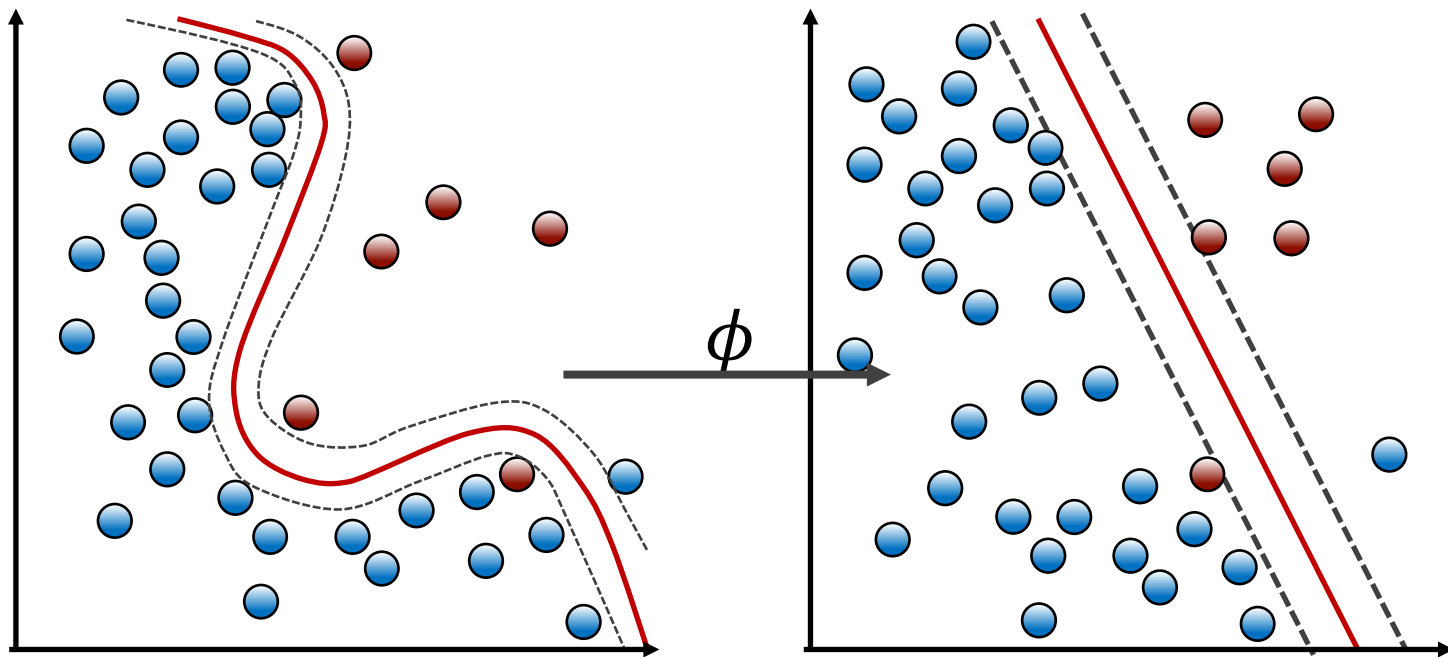
Kernels

Classification with SVMs



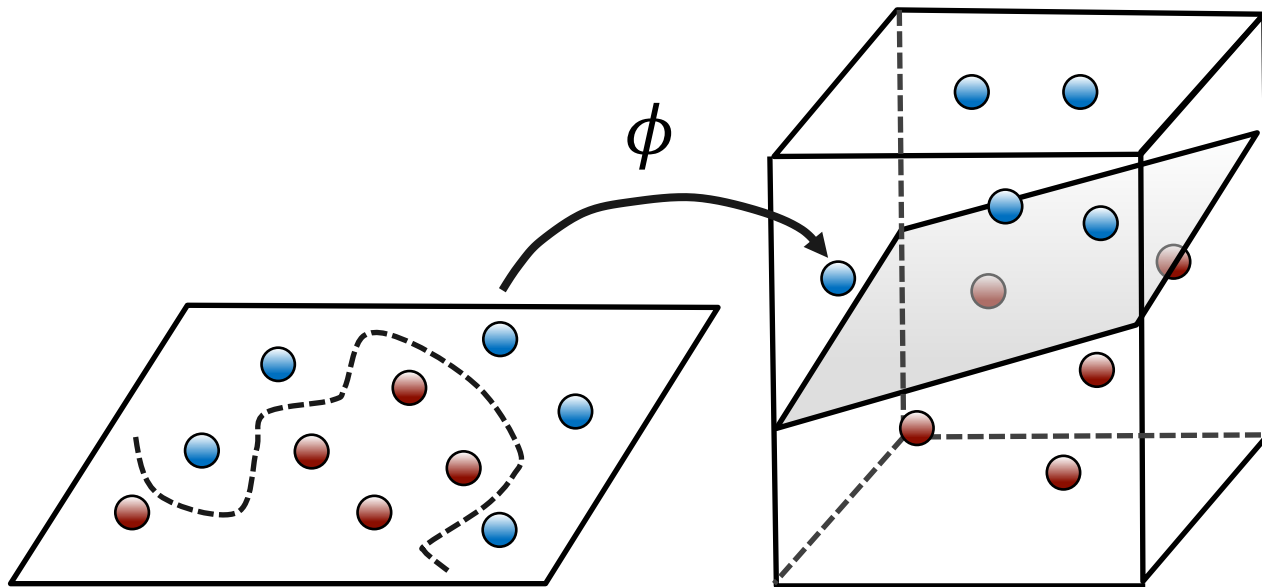
Non-Linear Decision Boundaries with SVM

Non-linear data can be made linear
with higher dimensionality



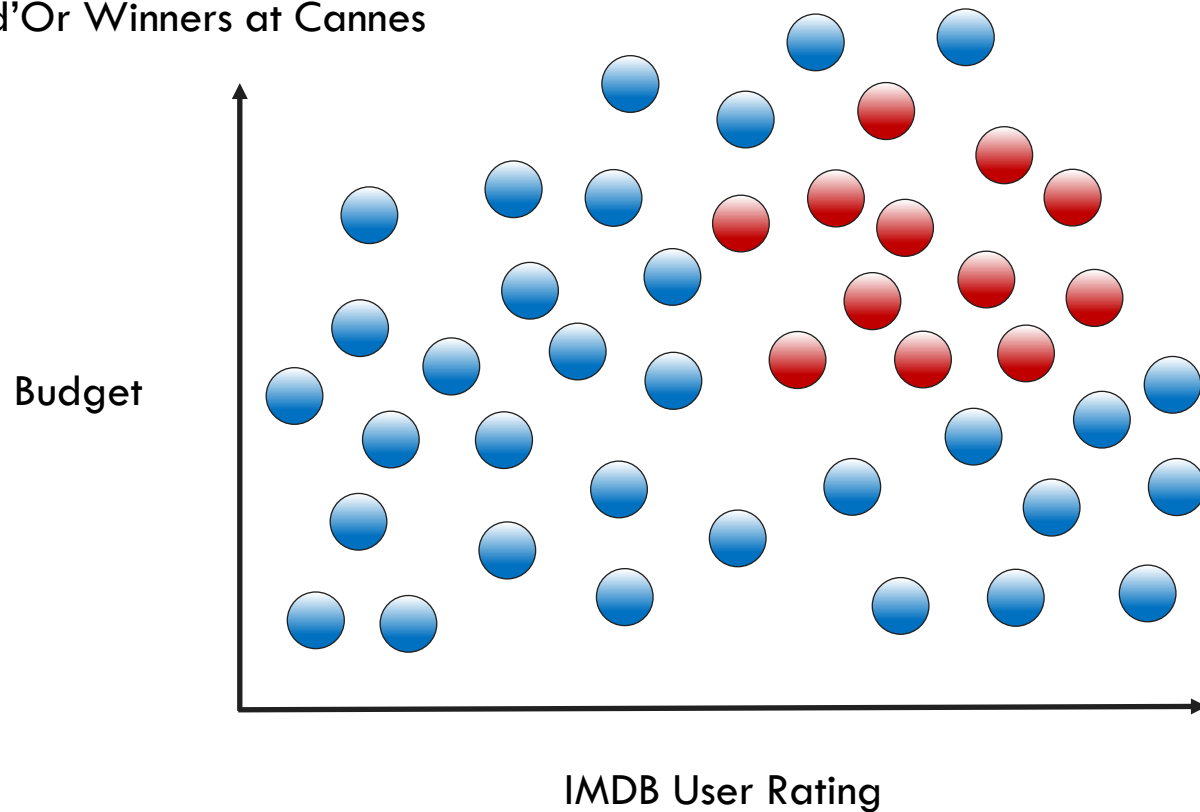
The Kernel Trick

Transform data so it is
linearly separable



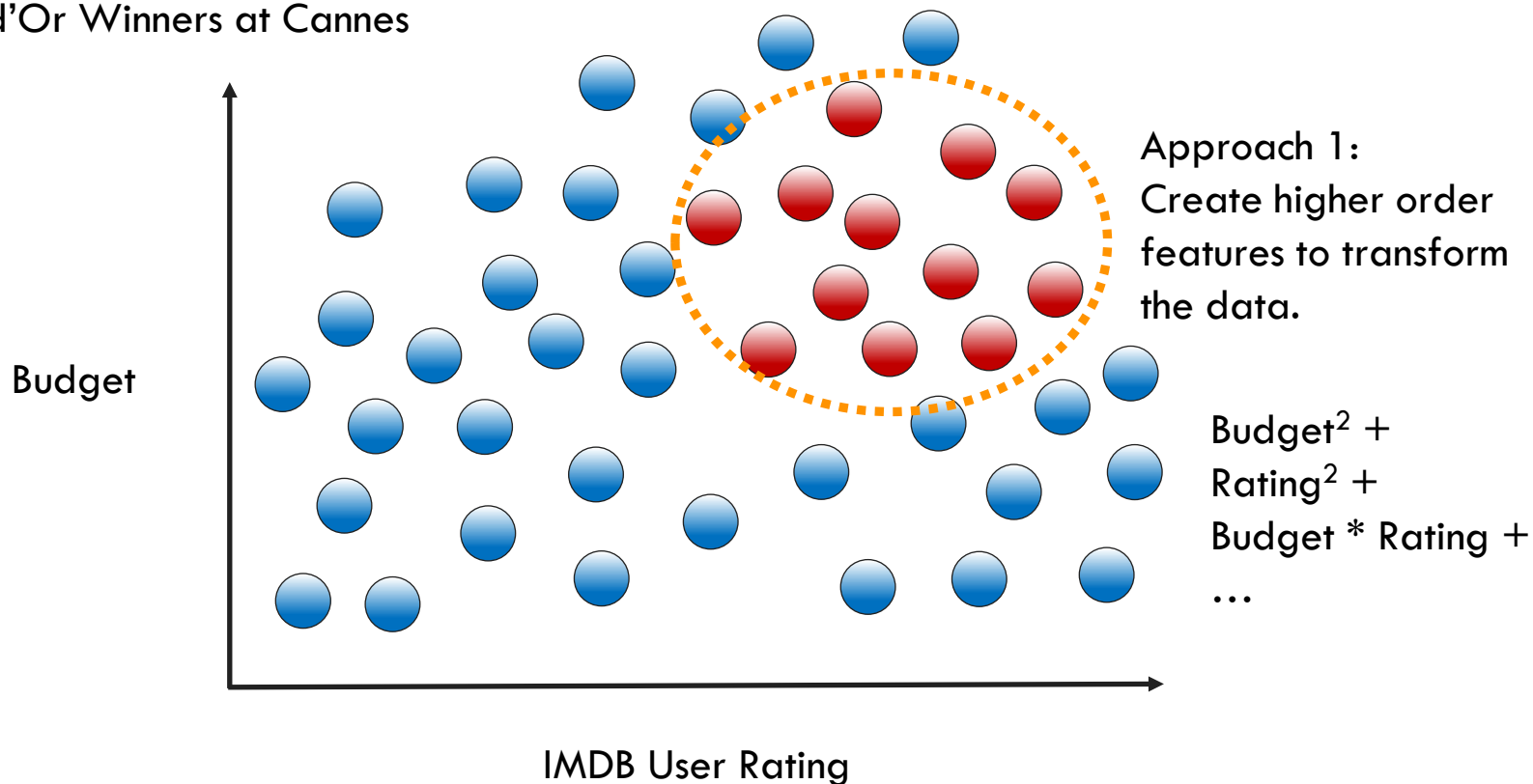
SVM Gaussian Kernel

Palme d'Or Winners at Cannes



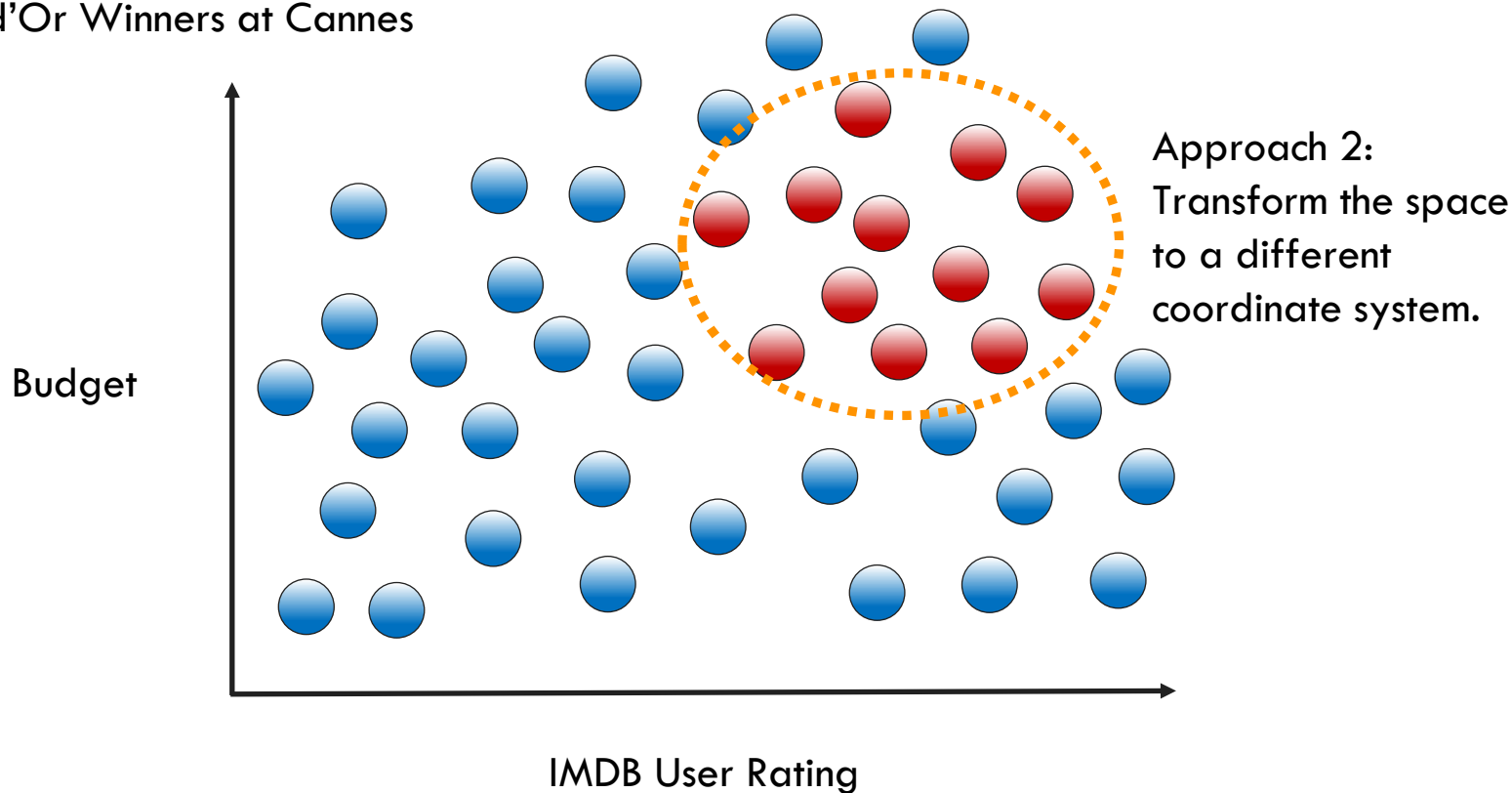
SVM Gaussian Kernel

Palme d'Or Winners at Cannes



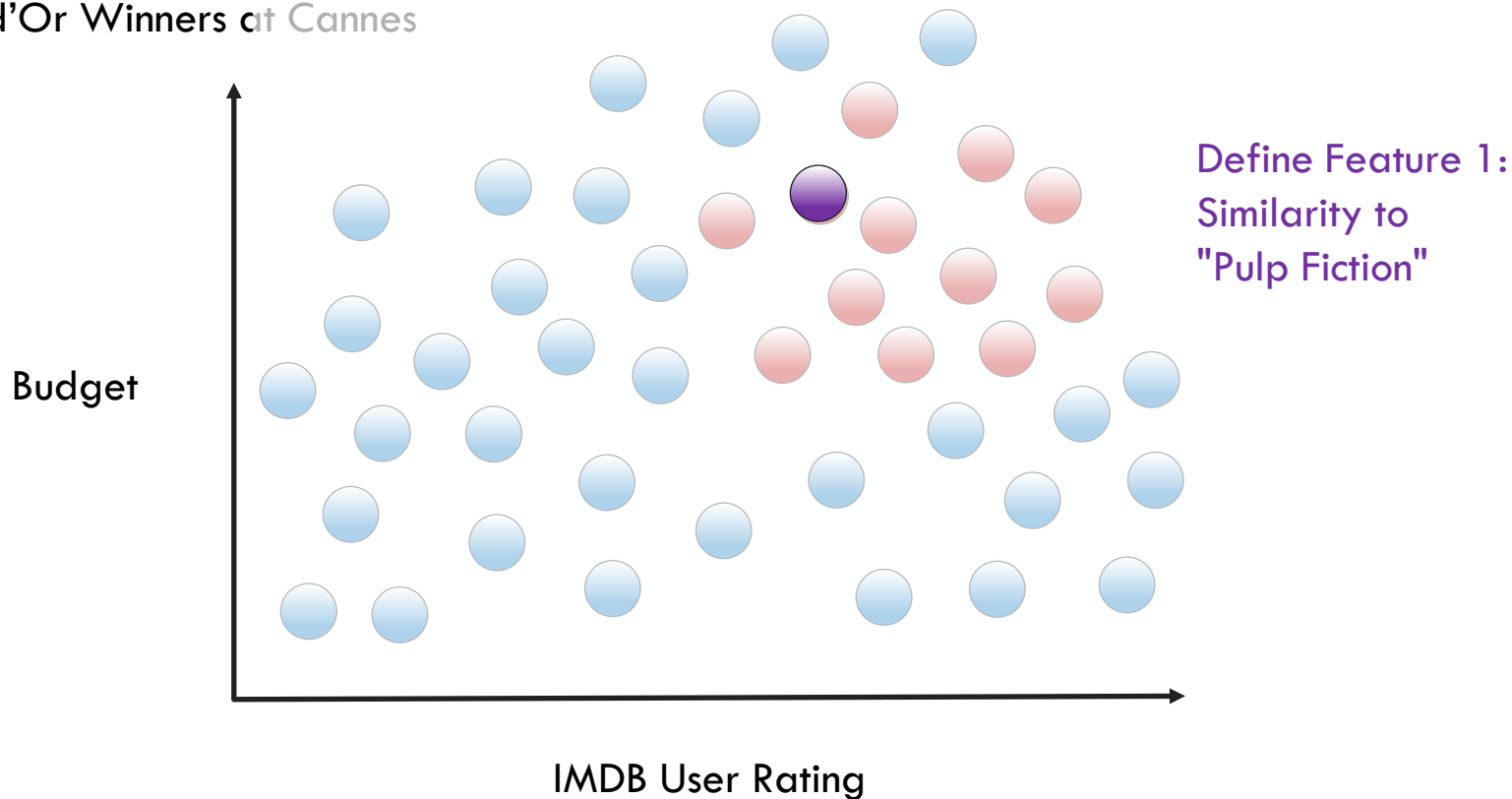
SVM Gaussian Kernel

Palme d'Or Winners at Cannes



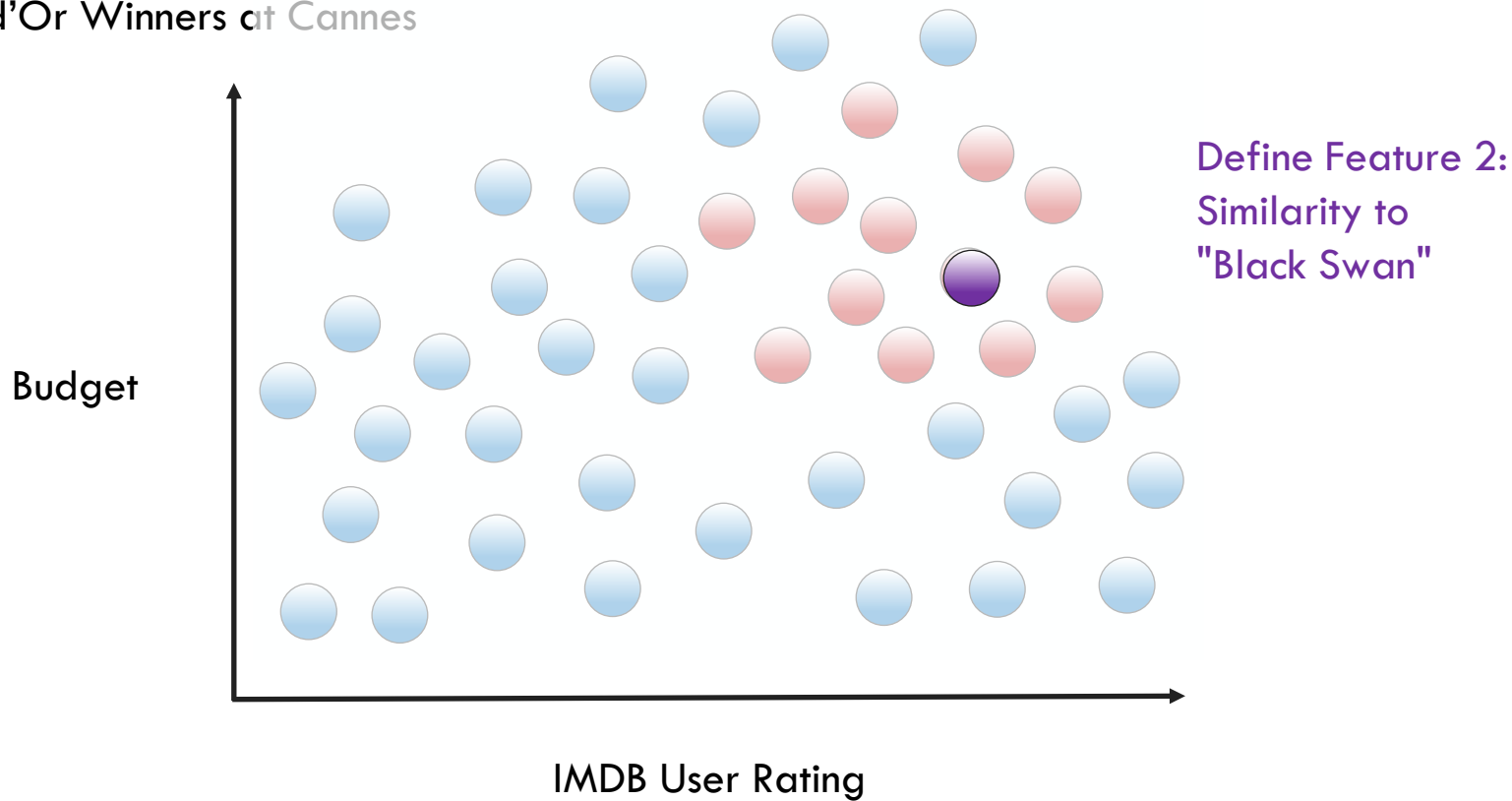
SVM Gaussian Kernel

Palme d'Or Winners at Cannes



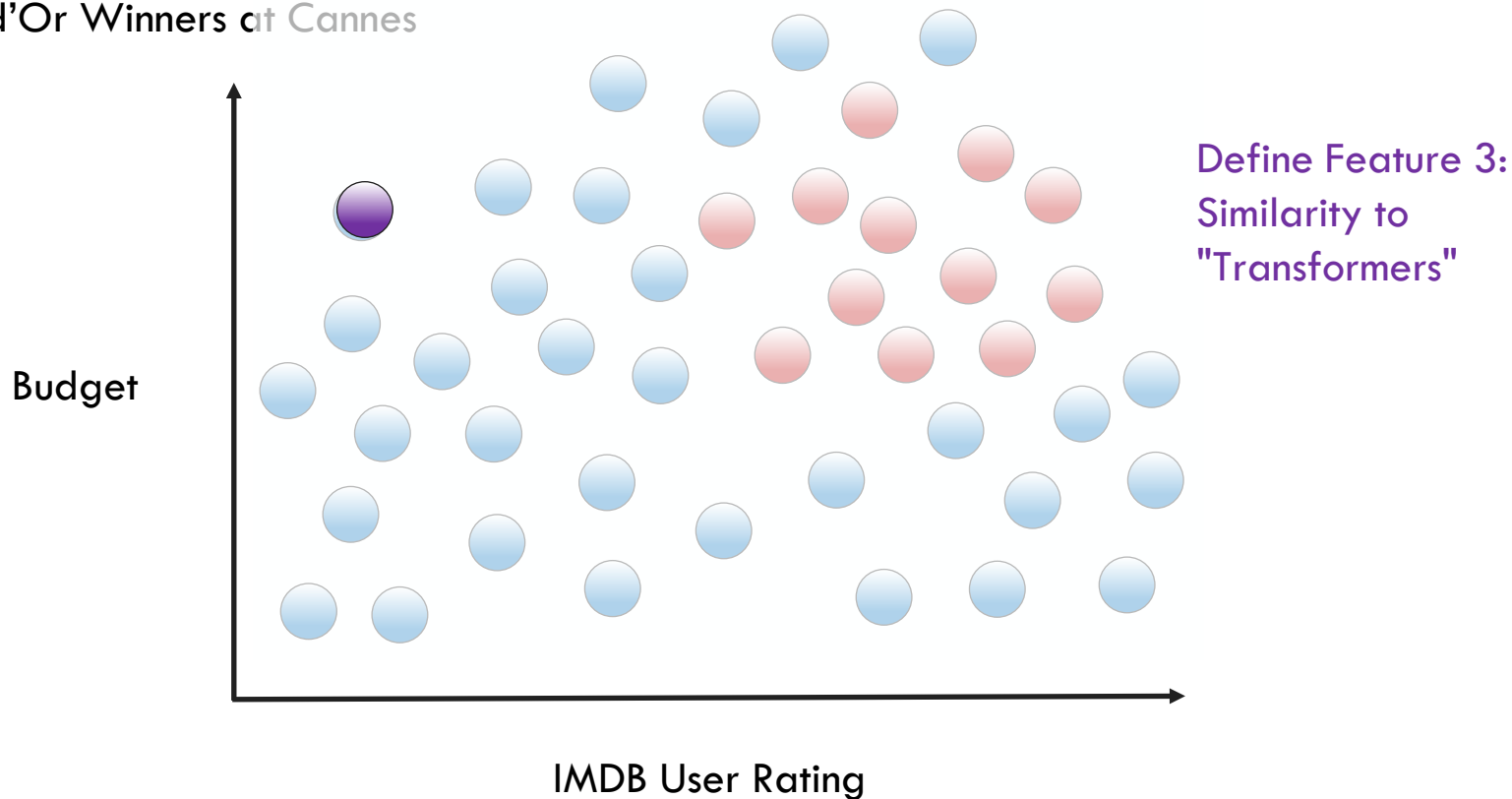
SVM Gaussian Kernel

Palme d'Or Winners at Cannes



SVM Gaussian Kernel

Palme d'Or Winners at Cannes

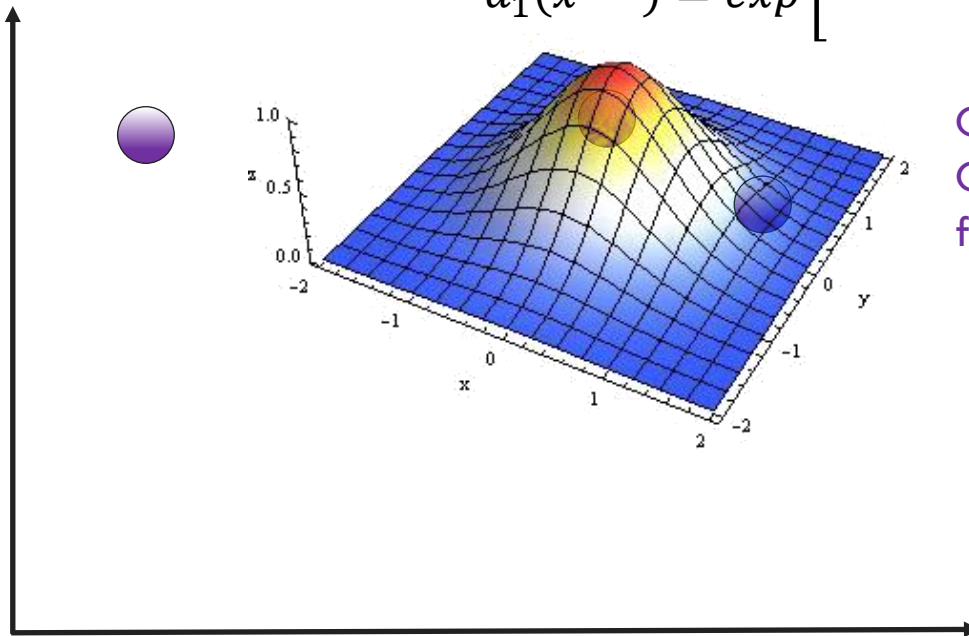


SVM Gaussian Kernel

Palme d'Or Winners at Cannes

$$a_1(x^{obs}) = \exp \left[\frac{-\sum (x_i^{obs} - x_i^{Pulp Fiction})^2}{2\sigma^2} \right]$$

Budget



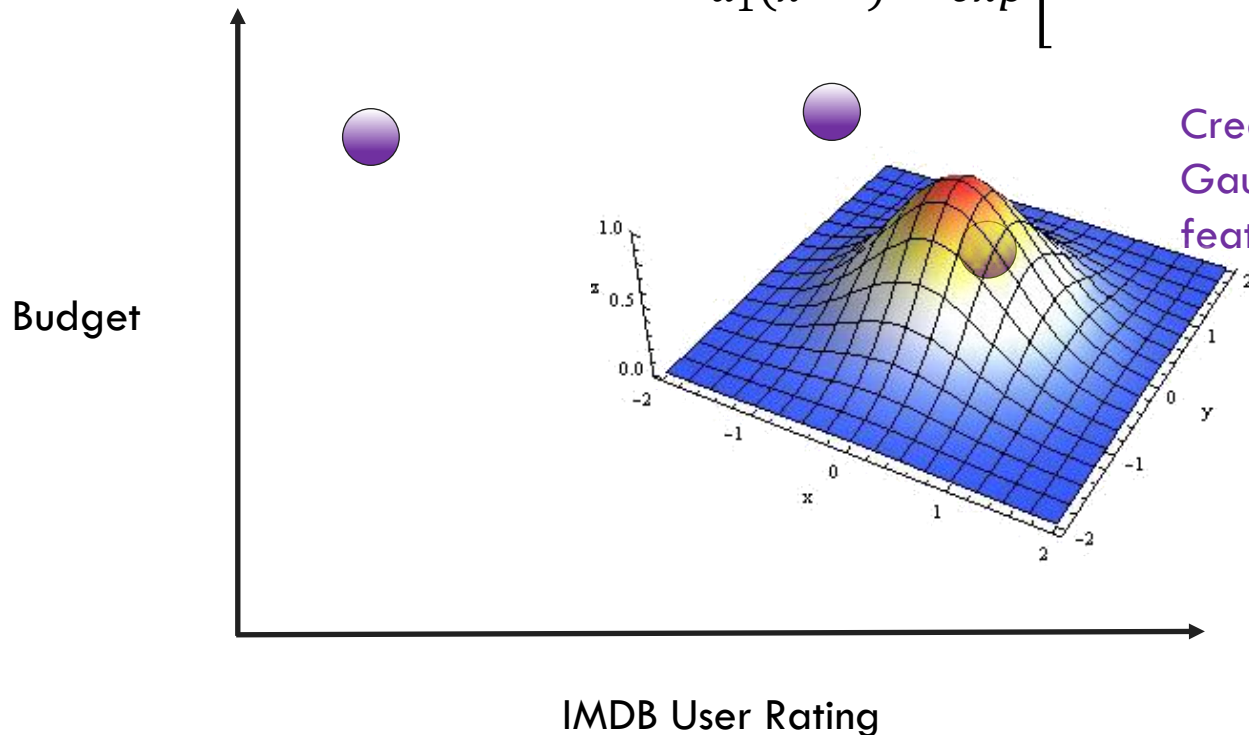
Create a
Gaussian function at
feature 1

IMDB User Rating

SVM Gaussian Kernel

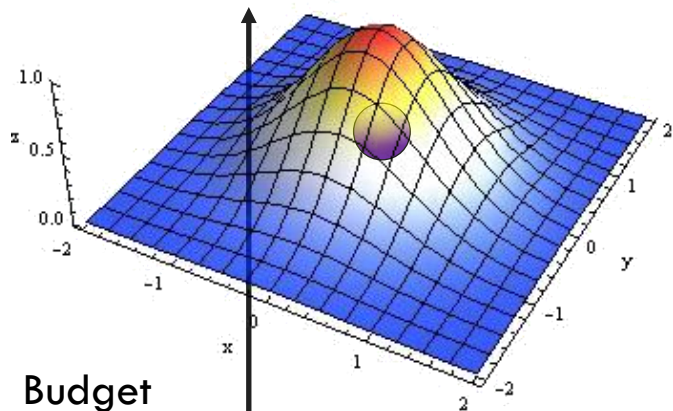
Palme d'Or Winners at Cannes

$$a_1(x^{obs}) = \exp \left[\frac{-\sum (x_i^{obs} - x_i^{Black\ Swan})^2}{2\sigma^2} \right]$$



SVM Gaussian Kernel

Palme d'Or Winners at Cannes



Budget

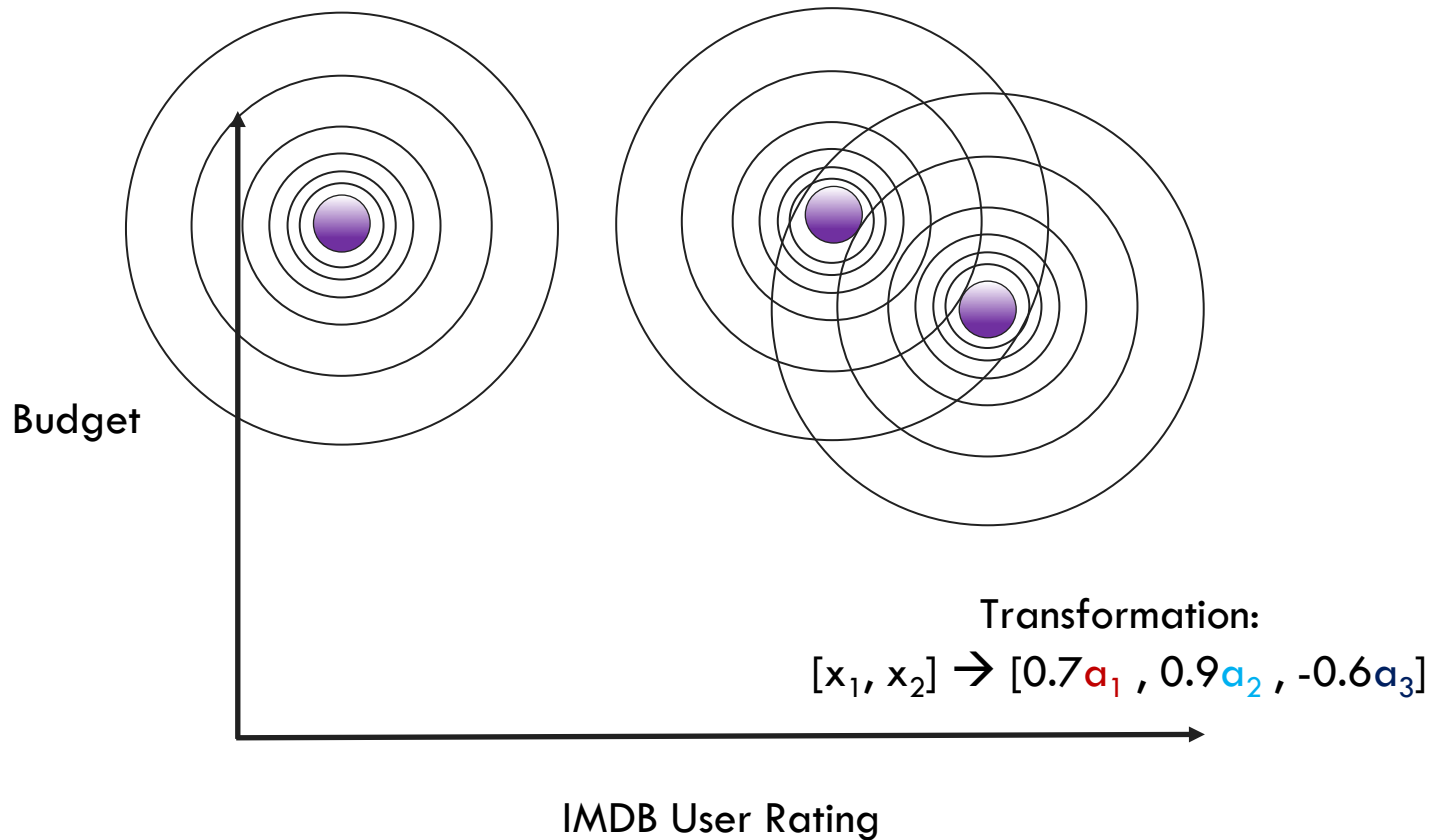
IMDB User Rating

$$a_1(x^{obs}) = \exp \left[\frac{-\sum (x_i^{obs} - x_i^{Transformers})^2}{2\sigma^2} \right]$$

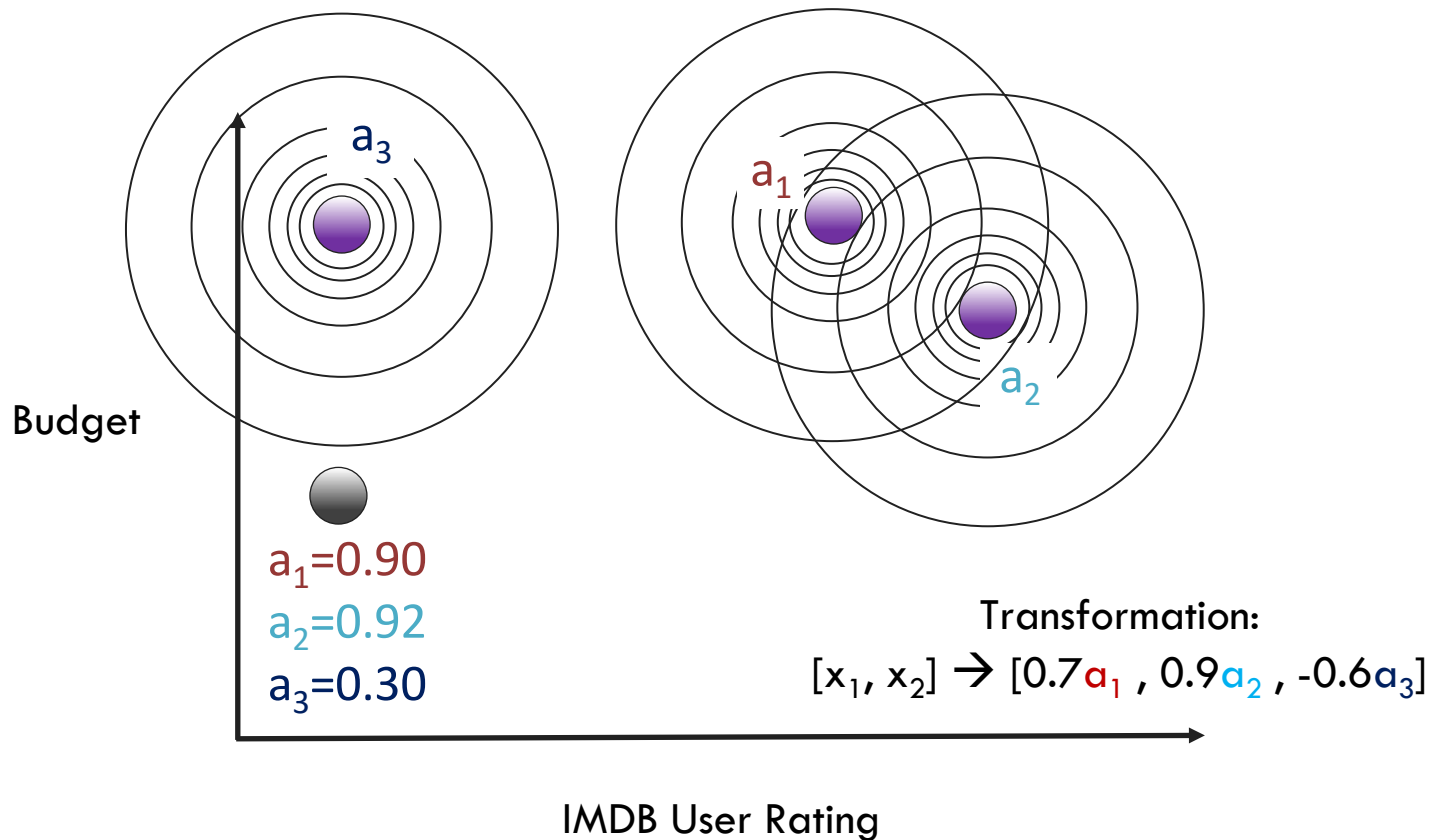


Create a
Gaussian function at
feature 3

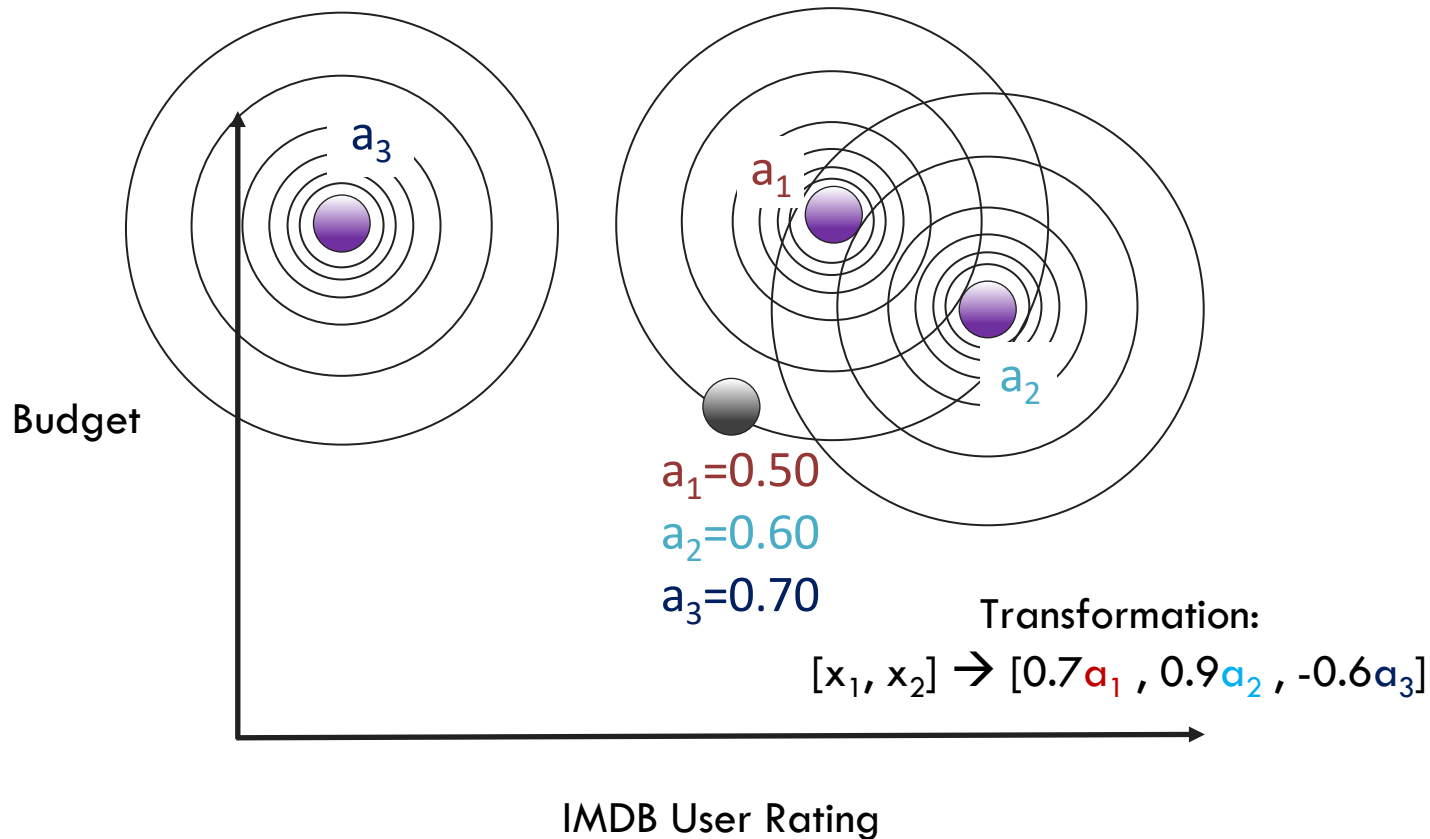
SVM Gaussian Kernel



SVM Gaussian Kernel



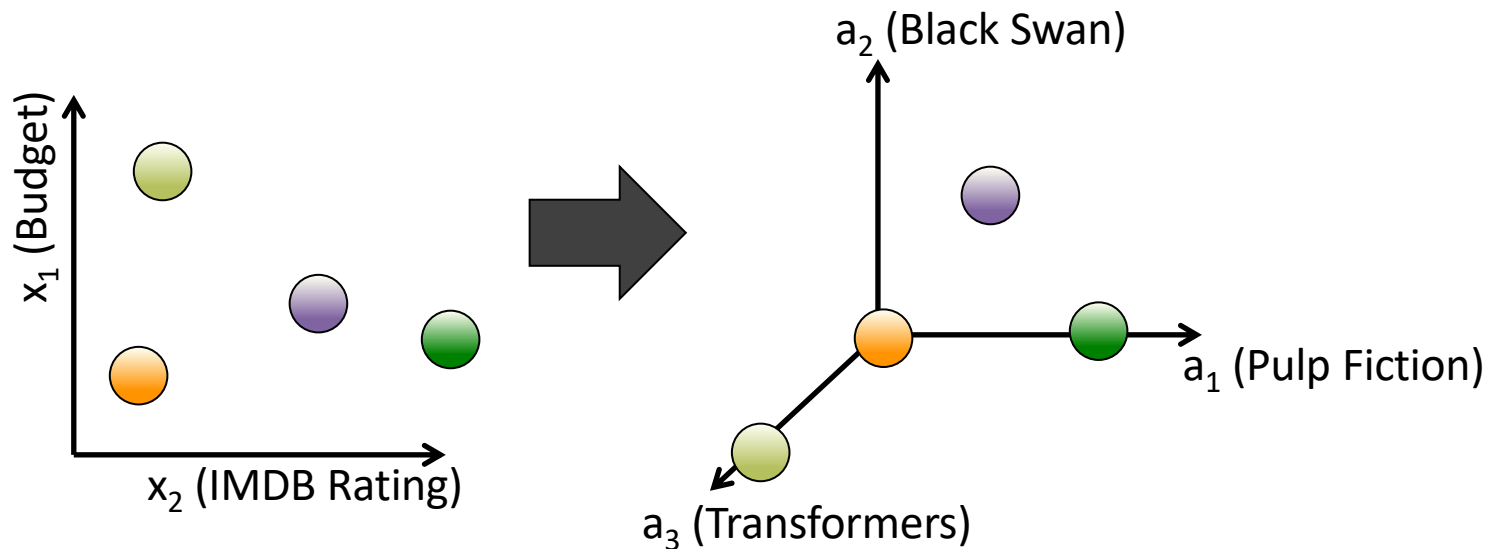
SVM Gaussian Kernel



SVM Gaussian Kernel

Transformation:

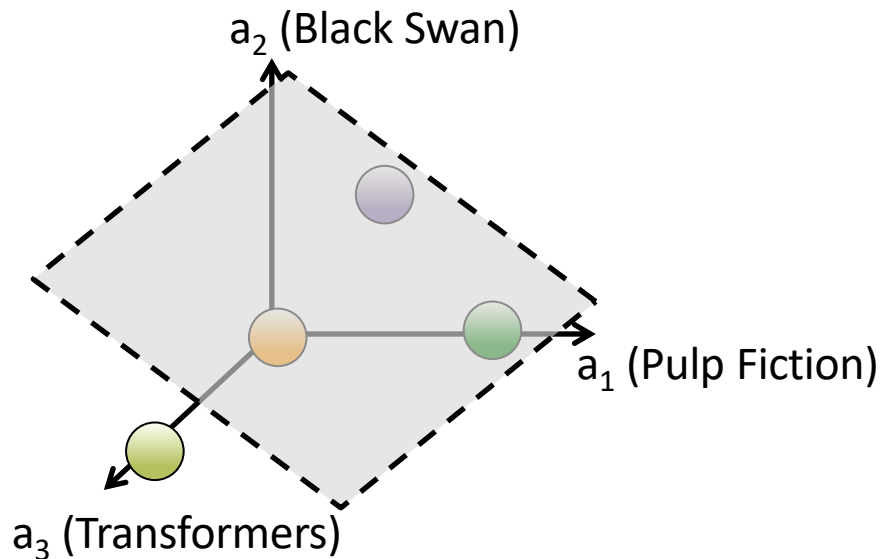
$$[x_1, x_2] \rightarrow [0.7\mathbf{a}_1, 0.9\mathbf{a}_2, -0.6\mathbf{a}_3]$$



Classification in the New Space

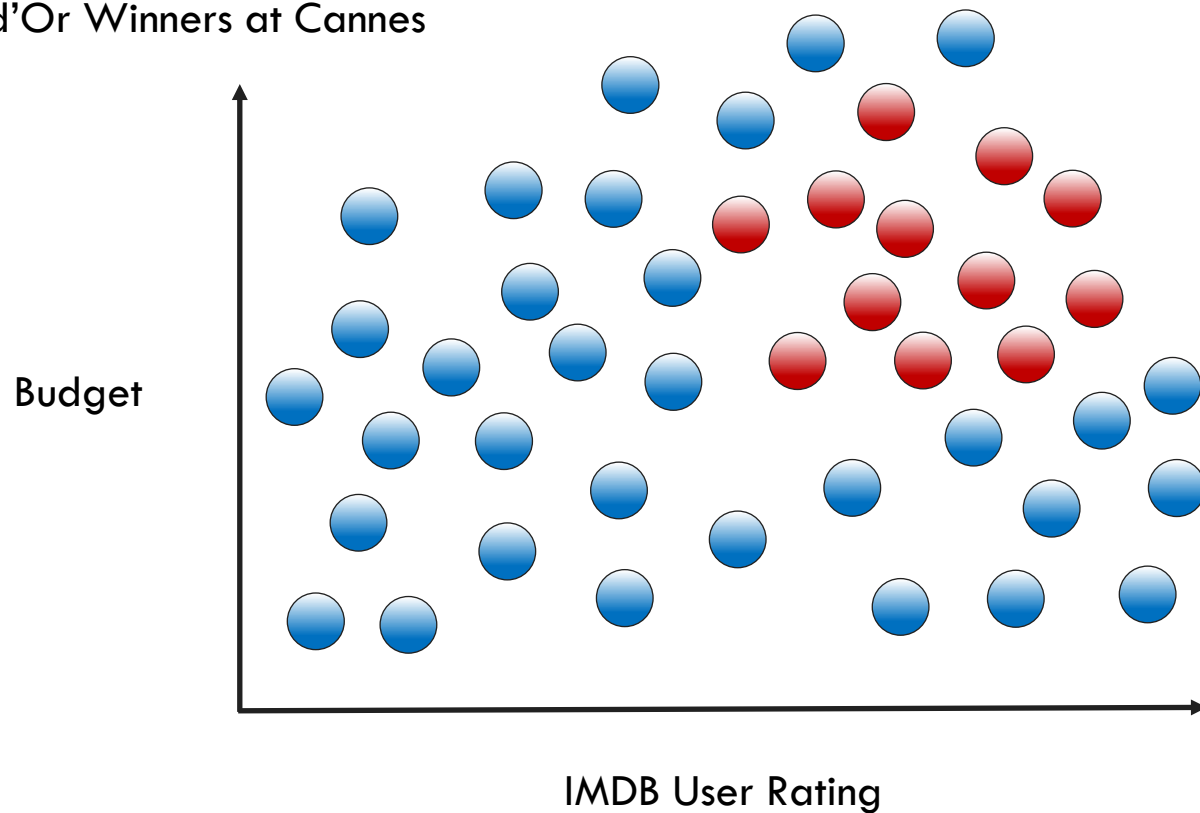
Transformation:

$$[x_1, x_2] \rightarrow [0.7\mathbf{a}_1, 0.9\mathbf{a}_2, -0.6\mathbf{a}_3]$$



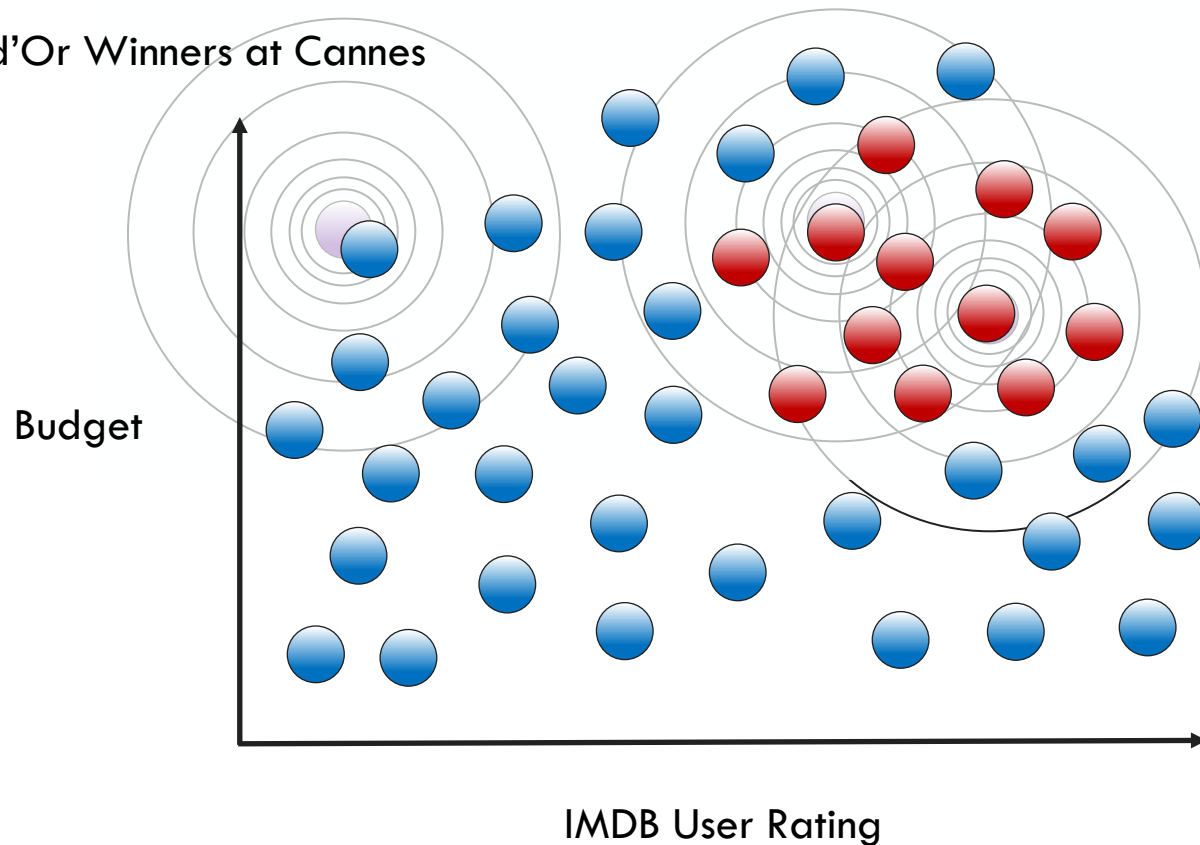
SVM Gaussian Kernel

Palme d'Or Winners at Cannes



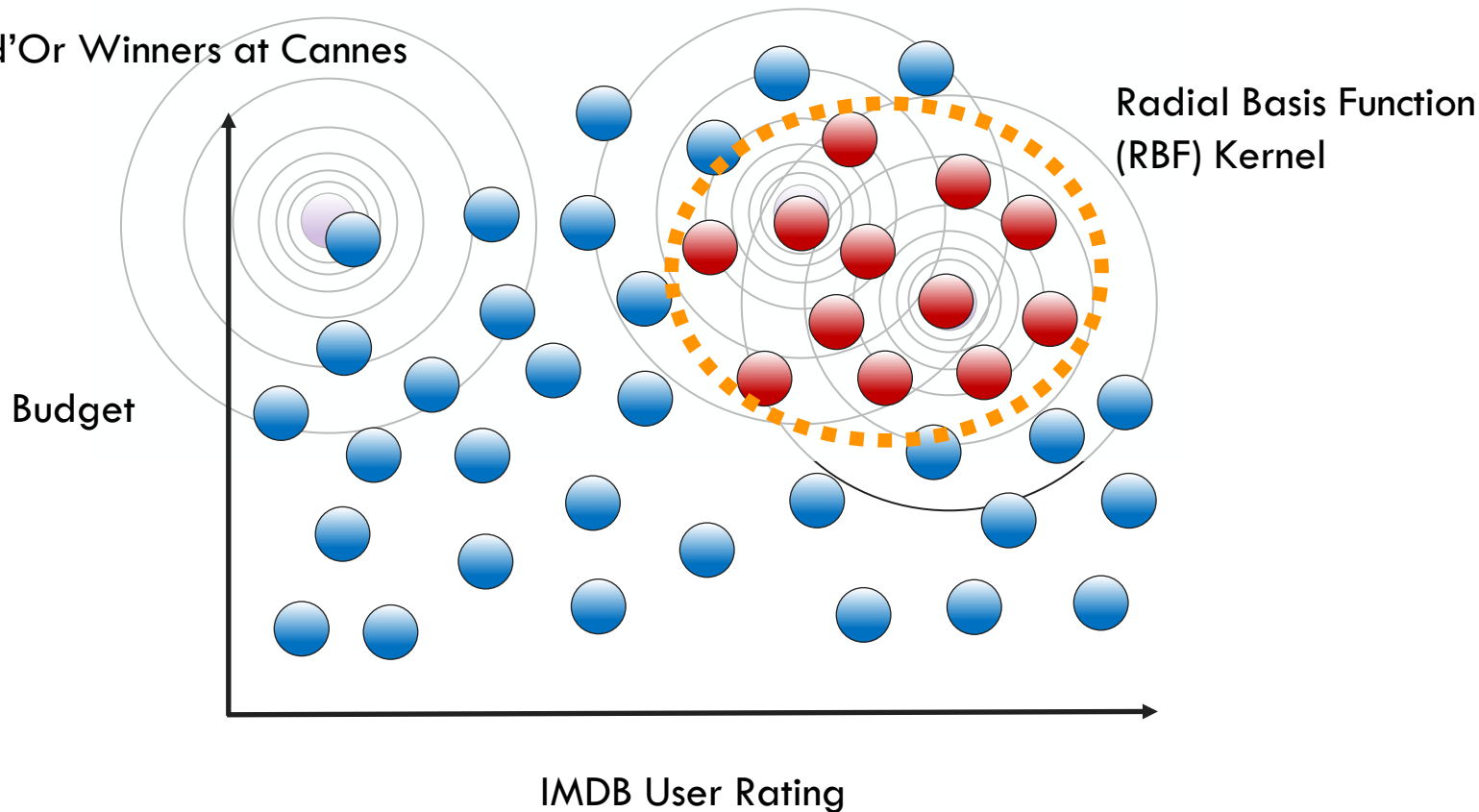
SVM Gaussian Kernel

Palme d'Or Winners at Cannes



SVM Gaussian Kernel

Palme d'Or Winners at Cannes



When to Use Logistic Regression vs SVC

Features

Data

Model Choice

Many (~10K Features)

Small (1K rows)

Simple, Logistic or LinearSVC

Few (<100 Features)

Medium (~10k rows)

SVC with RBF

Few (<100 Features)

Many (>100K Points)

Add features, Logistic, LinearSVC or
Kernel Approx.